# 10 ELECTRICITY & SOLAR

OUTLETS, FOUNTAINS, CAR CHARGER, PROPANE BACKUP

*Volunteers plan, dig and mount. Connections are made by Yusuf or a licensed electrician.*

| LEVEL | CREW | WHERE | STAGE NOW |
|---|---|---|---|
| ■■■ Licensed hands connect | 1–3 | Whole site; system lives in the kitchen shed | Load list + trench routes |

Hira runs on its own solar: panels on the kitchen shed, inverter and batteries inside. The next steps are outdoor outlets, solar fountains and a car charger. Volunteers can do a lot of this work (mapping, digging, mounting panels, building fountains), but never the live connections.

## FIRST STEPS *today*

1. Yusuf: write down the inverter model, battery capacity (kWh) and total panel watts. Pin it inside the shed.
2. Walk the site and flag every place an outlet is wanted: lounge, Lapis Room, dome, coop, container, fountains.
3. Write the load list for each spot.
4. Sketch trench routes from the kitchen shed to the flags, avoiding the dome ring, the pond and the wall line.
5. Build and test a standalone solar fountain kit.

## BEFORE YOU START

- **Volunteers never open the inverter or battery enclosure, or work on any wire that could be live.** Batteries deliver dangerous current even when the panels are covered.
- Every circuit is switched off and locked out before anyone touches it.
- Electrical work in unincorporated LA County generally needs a permit. Final tie-ins are made by Yusuf or a licensed electrician.
- Call 811 (DigAlert) before trenching.

## THE BASICS

**Load list** — Every device's watts × hours used per day = watt-hours per day. Add them up and compare with what the system produces and stores.

**Outdoor outlets** — GFCI-protected, with weatherproof "in-use" (bubble) covers.

**Burial depth** — Common figures under the electrical code: UF cable buried directly, 24 in; PVC conduit, 18 in; a 120V GFCI-protected residential circuit, 12 in. The electrician or inspector confirms. Lay warning tape about 12 in above the line.

**Solar fountains** — Easiest win: a standalone kit (a small panel wired straight to a DC pump), not connected to the main system. It runs while the sun shines; add a small battery for evenings. Not for the koi pond, which needs 24/7 oxygen.

**Sun for power, propane for heat** — With the propane tank arriving, put the biggest heat loads on gas: water heating, cooking, any space heating. Those would drain batteries fast; on propane, the solar is left for lights, outlets, pumps and the studio.

**Propane backup generator** — A propane (or dual-fuel) generator can top up the batteries during long cloudy stretches in winter, or carry a heavy load like car charging. It connects through the inverter's generator input: Yusuf or an electrician makes that connection.

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|---|---|
| **Car charging, honestly** | An EV goes about 3–4 miles per kWh. One 400W panel in LA makes about 1.6–2 kWh a day: roughly 5–7 miles. Level 1 (120V, ~1.4 kW) is slow but possible if the inverter and battery have headroom. Level 2 (240V, 7+ kW) likely exceeds the current system. A dedicated solar carport (6–10 panels) charging only the car also makes shade. |

## THE PLAN *in order*

| | | |
|---|---|---|
| 01 | **Inventory + load list** | What the system can do vs. what's wanted. |
| 02 | **Route plan** | Trench map approved by Yusuf or the electrician. |
| 03 | **Permit** | Pulled before trenching. |
| 04 | **Trench + conduit** | Volunteers dig and lay conduit and pull boxes. |
| 05 | **Connections** | Yusuf or electrician: wiring, GFCI outlets, breakers. Inspection. |
| 06 | **Fountains** | Standalone kits, installed by volunteers. |
| 07 | **Car charging** | Level 1 from the main system, or a dedicated solar carport array. |
| 08 | **Propane backup** | Generator sized and connected to the inverter's generator input; gas line from the same tank. |

## TOOLS & MATERIALS

Volunteers: shovels, rented trencher, PVC conduit, warning tape, flags, string

Qualified only: multimeter, crimpers, torque driver, lockout tags

### FOR YUSUF TO DECIDE

- ? Which outlets, where
- ? Level 1 or solar carport
- ? Future array size
- ? Fountain locations
- ? Propane backup generator: yes or no

**Resources:** DigAlert: call 811 (digalert.org) · LA County Public Works permits (EPIC-LA) (epicla.lacounty.gov)